AIRBUS A320 - TECHNICAL QUESTIONS

PRECISION APPROACH MINIMUMS

Category	Decision Height	RVR/Visibility
CAT I	not less than 200 ft	not less than 550 m (800 m visibility)
CAT II	lower than 200 ft and higher than 100 ft	not less than 300 m
CAT IIIa	lower than 100 ft or no DH	not less than 175 m
CAT IIIb	lower than 50 ft or no DH	less than 175 m but not less than 50 m
CAT IIIc	no DH	no RVR

COMMUNICATION LOSS

Try to troubleshoot, transmit in blind, squawk 7600 and use alternate means.
If in VMC, continue visual to the nearest suitable airport.
If in IMC, maintain the latest assigned altitude and speed for 20 minutes (non radar airspace), and then, follow the filed flight plan. Seven minutes (radar service airspace), maintain latest assigned speed and level or MSA, whichever is higher, then:

- Current - fly the current/assigned flight plan by the ATS, or
- Vectored - if being vectored fly the most direct manner to rejoin the filed flight plan
- Fly over the destination procedure fix and initiate the approach. Try to land within 30 minutes of the ETA

SPEEDS DEFINITIONS

The characteristic speeds on the PFD are computed by the FAC according to the FMS weight data.
On the Airbus, the stall speed is VS1g (due to the low-speed protection / alpha limit). As a result, V2 is 1.13 VS1g and VREF is 1.23 VS1g.

- VLS: lowest selectable speed. Represented by the top of an amber strip. Computed by the FAC, It's 1.13 VS1g during takeoff or touch and go. Becomes 1.23 VS1g after one step of flap retraction and 1.28 VS1g in clean configuration
- F: minimum speed the flaps may be retracted at takeoff. In approach, target speed in CONF 2 ou 3
- S: minimum speed the slats may be retracted after takeoff. In approach, target speed in CONF 1
- O: green dot speed is the engine-out operating speed in clean configuration (best lift-to-drag speed)

The limit speeds are:

- VA: maximum maneuvering speed. Full control deflection
- VMCG: during takeoff, with a failure of the critical engine, aircraft can be controlled by using primary flight controls with the remaining engine at takeoff thrust
- VMCA: in the air, the aircraft can be controlled with a bank of 5° during an engine failure with the remaining engine at takeoff thrust (takeoff flap setting and gear retracted)
- VMCL: in the air, the aircraft can be controlled with a bank of 5° during an engine failure with the remaining engine at TOGA thrust (approach flap setting)
- VFE: maximum speed for each flap configuration
- VLE: maximum speed with landing gear extended
- VLO: maximum speed for landing gear operation
- VMO: maximum speed

- VFE NEXT: maximum speed for the next flap lever position

The protection speeds are:

- VaPROT: angle of attack protection speed. The speed that the angle of attack protection becomes active. Represented by a black and amber strip on the PFD speed scale (normal law only)
- VaMAX: maximum angle of attack speed. The maximum angle of attack speed that may be reached in normal law. Represented by a red strip along the PFD speed scale (normal law only)
- VSW: stall warning speed. Represented by a red and black strip along the PDF speed scale when normal law is inoperative
- VMAX: represented by a black and red strip along the PFD speed scale. It's determined by the FAC. It's equal to VMO, VLE or VFE

Some other speeds:

- V1 is the highest speed the pilot can decide to continue or reject the takeoff
- VR is the speed the pilot rotates to reach V2 at 35 ft with an engine failure
- V2 is the takeoff safety speed, with OEI, to be reached at 35 ft and to be maintained during the second segment. It's equal to 1.13 VS1g
- VREF is the reference speed during normal final approach. Equal to VLS in CONF FULL
- VAPP is the final approach speed. Equal to VLS + wind correction (minimum of 5 kt and maximum of 15 kt)
-

RUNWAY DISTANCES DEFINITION

All the distances are provided by the airport authorities on the official publications (charts).

- TORA (Takeoff Run Available): is the length of the runway available for the takeoff run
- TODA (Takeoff Distance Available): is the TORA + clearway if any. A clearway is an area beyond the end of the opposite threshold not less than 152 m wide, maximum slope of 1,25 %, clear of obstacles and under control of the airport authorities. Takeoff performance is computed so the aircraft reaches 35 ft at V2 over the clearway
- ASDA (Accelerate and Stop Distance): is the TORA + stopway if any. Stopway is an area beyond the opposite runway threshold that is available for the aircraft, in case of a rejected takeoff, to come to a complete stop
- LDA (Landing Distance Available): is the length of the runway that is declared for the landing roll

RUNWAY CONDITIONS

- Dry: free of visible moisture and not contaminated. PERF: takeoff with no reversers
- Wet: the RWY is covered with visible moisture or water up to (including) 3 mm depth. PERF: takeoff with reversers. If the RWY is considered damp, the same applies
- Contaminated: hard contaminant (reduces friction forces) or a layer of fluid not considered as thin. PERF: takeoff with reversers, no flex temperature. The runway is covered and depth is over 3 mm with:
    - Dry snow: like a sand dust
    - Wet snow: snowball
    - Standing water
    - Slush: snow soaked with water
    - Compacted snow (hard contaminant)
    - Ice - cold and dry (hard contaminant)
    - Wet ice (hard contaminant)

TAKEOFF SEGMENTS

- 1st segment: from 35 ft (15 ft wet) at V2 until gear retraction
- 2nd segment: from gear retraction until OEI ACC ALT with a climb gradient of 2.4% at least

- 3rd segment: until the aircraft is accelerated to the final takeoff speed (green dot) and in clean configuration
- 4th segment: at final takeoff speed (green dot) with a minimum climb gradient of 1.2% and remaining engine at MCT

NADP 1 / NADP 2

Noise Abatement Departure Procedure.
NADP 1 reduces noise levels close to the airport.

- Speed: between V2+10 kt and V2+20 kt
- THR RED ALT: fixed at 800 ft
- At 3.000 ft, retract flaps / slats and accelerate to normal climb speed

NADP 2 reduces noise levels in areas distant from the airport.

- Speed: between V2+10 kt and V2+20 kt
- THR RED ALT: as takeoff performance
- At 800 ft, accelerate to green dot
- At 3.000 ft, accelerate to normal climb speed

TAKEOFF EMERGENCY BRIEFING

- Below 100 kt: consider to abort the takeoff if any ECAM (warning / caution) is activated
- Above 100 kt: “go-minded”. Very few situations may lead to reject the takeoff:
    - Engine fail / fire warning / severe damage
    - Unable / unsafe to fly
    - Any ECAM alert (Warning: APU FIRE, ENG FIRE, ENG OIL LO PR, F/CTL L+R ELEV FAULT, CONFIG / Caution: F/CTL SIDESTICK FAULT, ENG FAIL, ENG REVERSER FAULT, ENG REVERSER UNLOCKED, ENG 1(2) THR LEVER FAULT, FWS FWC 1+2 FAULT
    - If EGT exceedance or nose gear vibration, continue the takeoff
    - Tire failure / burst between V1-20 kt and V1, unless debris from the tires has caused serious engine anomalies, continue the takeoff

SAFETY CONDUCT OF THE FLIGHT

It’s considered impacted when:

- The flight crew has lost situational awareness, or
- The current pitch and thrust are not appropriate for the flight conditions, or
- The flight path is unexpected for the current flight conditions.

ENGINE FAILURE IN CRUISE

- THR levers: FLX / MCT
- A/THR: off
- HDG: as appropriate and pull if necessary
- Speed: green dot (obstacle strategy) or 0.78 / 300 kt (standard strategy) and pull. Fixed speed strategy is for ETOPS operation
- ALT: set the EO altitude and pull, PERF page for obstacle strategy and PROG page for standard strategy. When V/S is less than -500 ft/min, set V/S -500 ft/min and pull. Set the A/THR on
- If using the standard strategy, when level off, set speed manage

ENROUTE ALTITUDES

- Minimum Enroute Altitude (MEA): obstacle clearance (MOCA) and navaid reception (MRA). 1.000 ft for non-mountainous and 2000 ft for mountainous
- Minimum Obstacle Clearance Altitude (MOCA): obstacle clearance of 1.000 / 2.000 ft

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- Minimum Reception Altitude (MRA): an off-course navaid can be received for fix identification
- Minimum Crossing Altitude (MCA): altitude a fix must be crossed at or above due to terrain or signal coverage
- Minimum Off-Route Altitude (MORA): obstacle clearance off airways of 1.000 / 2.000 ft
- Grid MORA: obstacle clearance
- Minimum Sector Altitude (MSA): obstacle clearance of 1.000 ft within 25 nm radius of a navaid

FAC / ELAC / SEC

There are 2 FACs, 2 ELACs and 3 SECs.

- Flight Augmentation Computer has four main functions. Mnemonic FLY-W:
    - Flight envelope: PFD speed scale, MIN/MAX speed computation, VA computation and A.Floor protection
    - Low-energy aural alert: inhibited above 2.000 ft RA, below 100 ft RA, when TOGA or A.Floor active, GPWS triggered, both RA fault, Alternate or Direct law
    - Yaw damper: yaw damping, turn coordination, rudder trim and RTLU
    - Windshear detection: 3 s after liftoff until 1.300 ft RA during takeoff and from 1.300 ft RA until 50 ft RA and at least CONF 1

With the APs disengaged, each FAC interfaces with the ELACs or with FMGS with at least one AP on.
In normal conditions, FAC 1 controls yaw damper, turn coordination, rudder trim and RTLU. FAC 2 is on standby.
FAC 1 keeps aircraft within flight envelope through FD 1 and FAC 2 does the same through FD 2.

- Elevator Aileron Computer is used for normal elevator and stabilizer control, and aileron control. Consider 1 for normal law, 2 for alternate law and 3 for direct law
    - Normal, alternate and direct pitch (1-2-3)
    - Normal and direct roll (1-3)
    - Normal and alternate LAF (1-2)
    - Abnormal attitude
    - Aileron droop 5°
    - AP acquisition
- Spoilers Elevator Computer is used for normal spoiler control. It is on standby for elevator and stabilizer control
    - Alternate and direct pitch (2-3) - it's ELAC-1 (normal law)
    - Normal and direct roll (1-3) - same as ELAC
    - Alternate LAF (2) - it's ELAC-1 (normal law)
    - Abnormal attitude
    - Speed brake and ground spoilers

NORMAL LAW

It provides the following protections:

- Pitch attitude: 30° nose up in CONF 0 to 3 / 25° in CONF FULL (progressively reduced to 25° / 20° at low speed), 15° nose down
- Bank angle: it maintains 33° of bank with pitch autotrim. If the bank angle is beyond 33°, it returns to the 33° of bank angle. It limits the bank angle to 67°
- High speed: the aircraft recovers following a high speed upset (VMO + 16 kt/MMO + 0.04)
- Load factor: clean CONF from +2.5 until 1.0 g and in other CONF from +2.0 until 0 g
- High angle of attack: is activated when the AOA is greater than A.PROT. VaPROT, VaFloor and VaMAX are mainly computed based on the AOA (varies with CONF, weight and load factor)

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ALTERNATE LAW

Reduced protections:

- High speed stability: above VMO/MMO, a nose up demand is introduced to avoid an increase in speed
- Load factor
- Low speed stability replaces the High AOA protection. The A. Floor is inoperative. The VSW becomes present on the PFD speed scale

No reduced protections: only includes the load factor limitation.

DIRECT LAW

The pitch control is a direct stick-to-elevator relationship. The PFD displays “USE MAN PITCH TRIM”. No protections are operative.
The lateral control or roll direct law is a direct stick-to-surface-position relationship. It uses only aileron and spoilers 4 and 5. If spoiler 4 is faulty, spoiler 3 replaces it. If the aileron has failed, all spoilers become active.
The yaw mechanical control is so that the pilot controls yaw with the rudder pedals.

ABNORMAL ATTITUDE LAW

Is a specific set of control logics (pitch in alternate law, roll in direct law) that engages to assist the pilot in recovering when the aircraft exceeds extreme limits, such as a bank angle greater than 125° or extreme pitch attitudes (above 50° or below 30°).

MECHANICAL BACK-UP

It happens in a temporary total electrical loss, the temporary loss of five fly-by-wire computers, the loss of both elevators or the total loss of ailerons and spoilers.
The pilot manually applies trim to the THS to control the aircraft pitch. The PFD displays “MAN PITCH TRIM ONLY”.
The pilot uses the rudder pedals to laterally control the aircraft.

PRESSURIZATION

It has four functions:

- Ground function: fully opens the outflow valve on ground
- Prepressurization: during takeoff, increases cabin pressure to avoid a surge in cabin pressure during rotation
- Pressurization in flight: adjusts cabin altitude and rate of change
- Depressurization: after touchdown, gradually releases residual cabin overpressure before the ground function is active

The system consists of:

- 2 CPC (Cabin Pressure Controller)
- 1 RPCU (Residual Pressure Control) that automatically depressurizes the aircraft in case of abnormal residual pressure on ground when:
    - Outflow valve is not fully open; and
    - Both CPCs has failed or MAN mode is selected; and
    - ACFT is on ground; and
    - All ADIRS sense speed below 100 kt or both engines are shutdown
- 1 outflow valve on the right hand side of the fuselage behind the aft cargo compartment and below the flotation line
- 1 control panel
- 2 safety valves

The modes of operation are: ground (GN), takeoff (TO), climb (CL), cruise (CR), descent (DE) and abort (AB).

- Ground (GN): before takeoff and 55 s after landing, outflow valve fully open. At touchdown, rate -500 ft/min
- Takeoff (TO): pre-pressurizes to DeltaP of 0.1 psi (rate 400 ft/min)
- Climb (CL): automatically controls the rate of climb
- Cruise (CR): cabin altitude is at the level-off value or landing field elevation, whichever is higher, limited to 8000 ft
- Descent (DE): cabin pressure is equal to landing field elevation pressure +0.1 psi (max rate 750 ft/min)
- Abort (AB): controls the cabin altitude back to the altitude pressure of the takeoff airport when:
    - The aircraft does not climb more than 5000 ft AAL, or
    - The aircraft does not reach an altitude of 8000 ft MSL

PNEUMATIC

It supplies high-pressure air for:

- Air conditioning
- Engine starting
- Wing/engine anti-ice
- Water pressurization
- Hydraulic reservoir press
- Fuel tank inerting system (if installed)

There are three sources of high-pressure air:

- Engine bleed systems
- APU load compressor
- HP ground connection

It is controlled by two BMC (Bleed Monitoring Computer), the OVHD panel and the ECAM.
If one BMC fails, the other takes over control and monitors the system to warn on the ECAM if necessary:

- Overpressure
- Wing leak
- Pylon leak

The associated overtemperature protection is lost (the bleed valve does not close automatically) and the AIR APU BLEED LEAK alert is also lost if BMC 1 has failed. On the CEO variation only, the ENG BLEED LEAK warning is also lost.

DITCHING PB

If pressed:

- Closes the emergency ram air inlet
- Closes the avionics ventilation inlet and extract valves
- Closes the pack flow control valves
- Closes the outflow valve (only in CABIN PRESS AUTO mode)

ELECTRICAL

The minimum voltage to set the batteries to AUTO is 25.5 V. During normal operations, the minimum voltage is 26.5 V.
The battery check is to confirm that the current is below 60 A and decreasing within 10 s.
The batteries are connected to the DC BAT BUSS during APU start, battery charging (voltage below 26.5 V) and when AC BUS 1 and 2 are not powered and IAS less than 100 kt.
The electrical priority is (GEARB): ENG GEN, GPU GEN, APU GEN, RAT EMER GEN and batteries.
If the TR 1 fails, the TR 2 replaces it. DC ESS BUS is supplied via the ESS TR.

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If both TR 1 and TR 2 are lost, DC BUS 1, 2 and DC BAT BUS are lost. DC ESS BUS is supplied via the ESS TR.
The GEN 1 LINE push button on the EMER ELEC PWR overhead panel is used so that the GEN 1 supplies power to one fuel pump per tank. The AC BUS 1 is supplied by the GEN 2 through BTC. The GEN 1 BTC opens.

HYDRAULIC

It is composed of three independent systems named green, yellow and blue with normal pressure operation of 3.000 psi +/- 200 psi (2.500 psi on the blue system when powered by the RAT).
Power sources:

- Green: EDP 1, PTU as backup
- Yellow: EDP 2, PTU, electric and /hand pump as backup
- Blue: electric pump, RAT as backup

Normally, HP bleed air from ENG 1 pressurizes the hydraulic reservoirs. If air pressure is too low, the system takes bleed air pressure from the crossbleed duct.
The PTU works when the pressure differential between green and yellow systems is above 500 psi and for a self test during second engine start. It's inhibited when one engine master is off and parking brake set, during cargo door operation, NW STRG in the tow position and during first engine start.
If only the blue system is lost, CAT 3 DUAL, spoiler 3 and RAT EMER GEN are lost.

LAND ASAP

- LAND ASAP in amber: plan to land at the nearest suitable airport
- LAND ASAP in red: time-critical situation. Land immediately at the nearest airport

FAIL OPERATIONAL / FAIL PASSIVE

Both are automatic landing systems.
A failure that occurs below the alert height (AH: 100 ft) does not change the category of the system if fail operational.

- Fail operational (CAT 3 DUAL): in the event of a failure below AH, the system operates as a fail-passive system. If a failure that affects the redundant parts of the system or relevant ground equipment, a go around must be performed
- Fail passive (CAT 3 SINGLE): if a failure occurs, the pilot must take over control and there is no significant out-of-trim condition or deviation of the flight path / attitude

LVO / LVTO

- Low Visibility Operations: when RVR is less than 550 m and the ceiling is lower than 200 ft
- Low Visibility Takeoff Operations: when the RVR is below 400 m

WARNING / CAUTION MESSAGES

- Warning messages are red warnings (level 3) and require immediate action due to aircraft in dangerous configuration / limit flight conditions or a system failure altering flight safety
- Caution messages are amber (levels 2 and 1):
    - Level 2: the flight crew should be aware of the configuration or failure but no immediate action is required. These cautions should be considered without delay
    - Level 1: requires crew monitoring

READ AND DO PRINCIPLE

ECAM procedures, QRH procedures and OEB procedures.

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SUMMARIES

QRH procedures in case of ELEC EMER CONFIG or dual HYD failure.

- Situation assessment: ECAM actions, STATUS, summary (cruise), fuel consumption, landing performance. Decide / execute
- Approach preparation: STATUS, summary (approach, landing and go around), FMGS preparation
- Arrival briefing: summary (approach, landing and go around), STATUS
- Approach: summary (approach), STATUS

PREDICTIVE WINDSHEAR

When a PWS aural alert “WINDSHEAR AHEAD” or “GO AROUND WINDSHEAR AHEAD” is triggered, the flight crew must carefully check that there is no hazard, there is no signs of possible windshear and the reactive windshear system is operational.
The system operates when the aircraft is below 2300 ft RA and:

- The weather radar is on, or
- The weather radar is off and:
    - One engine running, and
    - GS greater than 30 kt, or
    - Longitudinal accel is above a threshold for at least 0,5 s

The system scans 5 NM ahead of the aircraft. If windshear is detected, a symbol appears on the PFD and ND (warning and caution). Advisory appears on ND only.
During takeoff, the system is inhibited between 100 kt and 50 ft RA. During approach, below 50 ft RA.

WINDSHEAR DETECTION FUNCTION

Provided by the FAC during takeoff and approach phase:

- At takeoff, 3 s after liftoff until 1.300 ft RA
- At landing, from 1.300 ft RA until 50 ft RA
- At least CONF 1 selected
-

RUNWAY LIGHTNING

- Runway centerline lights are white with a 50 ft distance between them. On the last 3.000 ft / 900 m, they become red and white. On the last 1.000 ft / 300 m, they become all red
- Runway edge lights are white with a 60 m distance between them. On the last 2.000 ft / 600 m, they become yellow

SELCAL

Selective calling is a feature that allows any ground radio station (like ATC) to alert a specific aircraft crew that they wish to communicate with them. Very used in remotes or oceanic areas. It allows the flight crew to keep the radio volume down.

FANS

Future Air Navigation System it’s an ICAO concept that uses digital data link and satellite-based navigation/surveillance to improve communication and reduce separation in oceanic and remote airspace.
FANS = CPDLC + ADS-C + GNSS.
On the A320, it’s available via the ATSU (Air Traffic Services Unit) and CMU (Communication Management Unit).

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ETOPS

Extended range Twin-engine Operational Performance Standards is a concept that requires you must always be within a certain flying time of a suitable diversion airport in case of engine failure or loss of systems (e.g.: electrical emergency configuration; loss of cockpit and cabin temperature control, it's recommended to open the cockpit door; loss of fuel pumps; engine fail or IDG fail; loss of air conditioning). Up to 60 minutes (ETOPS 60), no special approval is required. The A320neo is mostly common with the ETOPS 180, which means 180 minutes at OEI cruise speed (LRC).
On the A320, there is fuel x-feed check during the cockpit preparation.

ADIRS

It consists of the ADR (Air Data Reference) and the IR (Inertial Reference).
The ADR supplies:

- Airspeed
- Mach
- Overspeed warning
- Angle of attack
- Altitude
- Temperature

The IR supplies:

- Acceleration
- Angular rates
- Attitude
- Flight path vector
- HDG
- Track
- Ground speed
- Aircraft position

SRS

Speed Reference System is a managed vertical flight guidance mode available during takeoff and go-around.

- SRS Takeoff, the guidance law includes:
- FPA protection that ensures a minimum vertical speed of 120 ft/min
- A pitch angle protection to reduce the aircraft nose up (18º of maximum pitch attitude or 22.5º in case of windshear)
- When the aircraft is on the ground, V2 is the target speed
- When the aircraft is airborne, V2+10 kt is the target speed
- If one engine fails, the target speed is the current aircraft speed limited between V2 and V2+15 kt
- To engage the SRS TO mode:
    - The aircraft is on the ground for more than 30 s
    - The slats or flaps are extended
    - V2 is inserted in the PERF TAKEOFF page (if not, V/S mode engages 5 s after lift-off on the current V/S value)
    - At least one thrust lever at TOGA detent or FLX / MCT when FLX TO is entered in the PERF TAKEOFF page
- It disengages when:
    - Automatically at the ACCEL ALT with AEO (it does not automatically disengage if OEI), or if ALT* or ALT CST* mode engages, or
    - Another vertical speed mode engages

    - It reverts to OP CLB if the FCU selected altitude is above the aircraft altitude and the flight crew pulls the SPD / MACH knob
- The SRS GA mode guidance law includes:
    - A FPA protection that ensures a positive climb
    - A pitch angle protection to reduce the aircraft nose-up (18º or 22.5º in case of windshear)
    - A speed protection the ensures the aircraft does not exceed VFE
    - The speed target is the memorized aircraft speed at SRS GA engagement or VAPP, whichever is higher. The limit is the lowest value between VLS+25 kt (VLS+15 kt if OEI) and VMAX-5 kt
    - To engage SRS GA mode:
        - At least one thrust lever in TOGA
        - Flaps lever is at least in position 1
        - The aircraft is airborne or on the ground for less than 30 s
    - It disengages when:
        - At GA ACCEL ALT, or if ALT* or ALT CST* engages (above 400 ft RA)
        - The flight crew engages another vertical mode
        - It reverts to OP CLB if the FCU selected altitude is above the aircraft altitude and the flight crew selects a speed
        - If the AP/FD TCAS mode engages upon a RA

GO AROUND / DISCONTINUE THE APPROACH

Perform a go around or discontinue the approach when the aircraft altitude is above the FCU ALT.
To cancel an approach:

- Announce: “cancel approach”
- Press APPR ou LOC pb
- Select lateral, vertical modes and select speed and adjust

When the aircraft altitude is below the FCU ALT, a go around is mandatory.

MINIMUM FUEL / MAYDAY FUEL

- Minimum fuel: the pilot cannot accept any undue delay. The pilot is committed to landing at a specific aerodrome. The aircraft is landing with 30 minutes (reserve fuel) on board
- Mayday fuel: the aircraft will land with less than 30 minutes (reserve fuel) on board

ENG FIRE PUSH BUTTON

When pressed:

- Silences the aural alert
- Arms the squibs
- Closes the hydraulic fire shut off valve
- Closes the fuel low-pressure valve
- Deactivates the IDG
- Cuts off the FADEC
- Closes the engine bleed valve
- Closes the pack flow control valve

FADEC

During manual start, the FADEC aborts the engine start if:

- The start EGT limit is exceeded before the N2 reaches 50%, or
- Starter shaft shear fault occurs, or
- Dual N2 failure occurs

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ENGINE START

When the ENG MASTER ON:

- The LP fuel valve and engine start valve open
- When N2 is more than 15 % the ignition starts (on ground). It starts immediately in flight
- When N2 is more than 20 % the HP fuel valve opens
- When N2 is more than 55 % the ignition stops
- When N2 is more than 63 % the engine start valve closes

During automatic start on ground, only one igniter is supplied. During manual start, both igniters are supplied. If the first attempt fails, the second attempt uses the same igniter. If it still fails, then both igniters are used for the third attempt.

ENGINE MANUAL START

It's recommended when:

- After aborting a start because of (SEL):
    - Stall
    - EGT overlimit
    - Low air pressure
- When expecting (DDDI):
    - Degraded bleed performance
    - Degraded external pneumatic
    - Degraded engine EGT limit
    - Intermittent ENG 1(2) IGN A(B) FAULT

The FADEC has authority over (SHI):

- Start valve (opening and closing)
- High pressure fuel valve
- Igniters